Mathematical Statistics

Analysis of Variance

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Oneway Analysis of Variance

- So far, we modeled a random variable with a pdf or pmf that depended on parameters.
- We are interested in discussing the situation where a random variable can be modeled not only with unknown parameters but also with known covariates.
- We describe the methodologies of analysis of variance (ANOVA) and regression analysis to do so, which assumes a linear relationship.

Oneway Analysis of Variance

- The ANOVA is a method of estimating the means of several populations, where populations are assumed to follow normal distributions.
- Heart of the ANOVA is also the topic of statistical design. How can we get the most information on the most populations with the fewest observations?
- The ANOVA compares means among treatment groups, without assuming any parametric relationships (regression does assume such a relationship).
- A basic idea of the ANOVA : partition of variation in means, not the analysis of variance.

Oneway ANOVA

- In the oneway ANOV, we assume that data, Y_{ij}, are observed according to a model

$$Y_{ij} = \theta_i + \epsilon_{ij},$$

  where $i = 1, \ldots, I$, and $j = 1, \ldots, n_i$.

- Here the θ_i are unknown parameters and the ϵ_{ij} are error random variables. We do not assume equal numbers of observations in each treatment group.
- Without loss of generality, we assume that $E(\epsilon_{ij}) = 0$. Hence, $E(Y_{ij}) = \theta_i$.

Alternative Model

- An alternative (but overparameterized) and more interpretable model:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij},$$

  where $E(\epsilon_{ij}) = 0$.

- In this alternative model, μ can be thought of as a grand mean, the common mean level of the treatment, and τ_i is the unique effect due to treatment i, the derivation from the mean level that is caused by the treatment.

- However, we can not estimate both μ and τ_i, because there are problems with identifiability.

Identifiability

- A parameter θ for a family of distributions $\{f(x \mid \theta) : \theta \in \Theta\}$ is identifiable if distinct values of θ correspond to distinct pdfs or pmfs. That is, if $\theta \neq \theta'$, then $f(x \mid \theta)$ is not the same function of x as $f(x \mid \theta')$.
- If the model is not identifiable, then, there is a difficulty in parameter estimation.
- If there is an identifiability problem, we may solve this problem by redefining the model.
- Note the first model is identifiable, while the second model is not identifiable.
- In the second model, we have $I + 1$ parameters, but we have only I means, $E(Y_{ij})$. Without any further restriction on the parameters, more than more set of $(\mu, \tau_1, \ldots, \tau_I)$ will lead to the same distribution. In this case, it is common to add the restriction that $\sum_{i=1}^{I} \tau_i = 0$

Model and Distribution Assumption

Oneway ANOVA assumptions:

- Random variables Y_{ij} are observed with the following model:

$$Y_{ij} = \theta_i + \epsilon_{ij}, i = 1, \ldots, I, j = 1, \ldots, n_i,$$

where

1. $E(\epsilon_{ij}) = 0,\ Var(\epsilon_{ij}) = \sigma_i^2 < \infty.\ Cov(\epsilon_{ij}, \epsilon_{i'j'}) = 0$ if $i \neq i'$ or $j \neq j'$.
2. The ϵ_{ij} are independent and normally distributed, and $\sigma_i = \sigma$ for all i (homosceasticity)

Oneway ANOVA model

- Without assumption (ii) we could perform point estimation and possibly look for estimators that minimize variance within a class, but we could not do interval estimation. If we assume some distribution other than normal, intervals and tests can be difficult. Of course, with reasonable sample size and populations that are not too asymmetric, we can use the Central Limit Theorem (CLT).
- The equality of variance assumption is also important.

Equality of Variance Assumption

- When the ANOVA assumption is violated, we may try to transform the data to make the ANOVA assumption true.
- One common transformation, $Box-Cox$ transformation family;

$$g_\lambda(y) = (y^\lambda - 1)/\lambda \text{ if } \lambda \neq 0, g_\lambda(y) = log(y) \text{ if } \lambda = 0.$$

- Box (1964) shows that the robustness of the ANOVA to the assumption of normality depends on how equal the variances are (equal being better)

Classic ANOVA Hypothesis

- $H_0 : \theta_1 = \ldots = \theta_I$ versus H_1: one of $\theta_i' s$ is not different.
- Let $Y_{ij} =$ the jth observation of the ith treatment .
- The analysis of variance is based on the following identify:

$$SS_{TOT} = \sum_{i=1}^{I}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{..})^2 = \sum_{i=1}^{I}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i.})^2 +$$

$$\sum_{i=1}^{I} n_i(\bar{Y}_{i.} - \bar{Y}_{..})^2,$$

where $\bar{Y}_{i.} = \frac{1}{I}\sum_{j=1}^{n_i} Y_{ij}$, the average of the observations under the ith treatment, and $\bar{Y}_{..} = \frac{1}{\sum_i n_i}\sum_{i=1}^{I}\sum_{j=1}^{n_i} Y_{ij}$, the overall average.

Classic ANOVA Hypothesis

- Let

$$SS_W = \sum_{i=1}^{I} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2$$

and

$$SS_B = \sum_{i=1}^{I} n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2$$

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$$SS_{TOT} = SS_W + SS_B.$$

That is, the total sum of squares equals to the sum of squares within groups plus the sum of squares between groups.

- The term SS_W reflects the variation of the data within the treatment groups, and SS_B reflects variation of the treatment means among or between treatments.

Theorem A

- Under the model, $Y_{ij} = \theta_i + \epsilon_{ij}$, if the errors, ϵ_{ij}, are independent with means 0 and common variances σ^2,

$$E(SS_W) = \sum_{i=1}^{I}(n_i - 1)\sigma^2,$$

and

$$E(SS_B) = \sum_{i=1}^{I} n_i E(\bar{Y}_{i.} - \bar{Y}_{..})^2 =$$

$$= \sum_{i=1}^{I} n_i(\theta_i - \bar{\theta})^2 + (I-1)\sigma^2$$

where $\bar{\theta} = \sum_i \theta_i / I$.

Proof of Theorem A

- We will use the following lemma: If X_i's are independent with $E(X_i) = \mu_i$ and $Var(X_i) = \sigma^2$, then

$$E(X_i - \bar{X})^2 = (\mu_i - \bar{\mu})^2 + \frac{n-1}{n}\sigma^2,$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

- Since $E(Y_{ij}) = \mu + \tau_i$ and $Var(Y_{ij}) = \sigma^2$, applying the lemma with the role of X_i being played by Y_{ij} and that of $\bar{X}$ being played by $\bar{Y}_{i\cdot}$, we obtain the following

$$E(SS_W) = \sum_{i=1}^{I}\sum_{j=1}^{n_i} E(Y_{ij} - \bar{Y}_{i\cdot})^2$$

$$= \sum_{i=1}^{I}\sum_{j=1}^{n_i}[(\theta_i - \theta_i)^2 + \frac{n_i - 1}{n_i}\sigma^2] = \sum_{i=1}^{I}\sum_{j=1}^{n_i}\frac{n_i - 1}{n_i}\sigma^2$$

$$= \sum_{i=1}^{I}(n_i - 1)\sigma^2.$$

Proof of Theorem A, continued

- Since $E(\bar{Y}_{i.}) = \theta_i$ and $Var(\bar{Y}_{i.}) = \sigma^2/n_i$, applying the lemma with the role of X_i being played by $\bar{Y}_{i.}$ and that of $\bar{X}$ being played by $\bar{Y}_{..}$, we obtain the following

$$E(SS_B) = \sum_{i=1}^{I} n_i E(\bar{Y}_{i.} - \bar{Y}_{..})^2$$

$$= \sum_{i=1}^{I} n_i [(\theta_i - \sum_{j=1}^{I} \theta_j/I)^2 + \frac{I-1}{I}\sigma^2/n_i] =$$

$$= \sum_{i=1}^{I} n_i(\theta_i - \bar{\theta})^2 + (I-1)\sigma^2,$$

where $\bar{\theta} = \sum_{i=1}^{I} \theta_i/I$.

Estimation of Variance

- SS_W may be used to estimate σ^2; that is,

$$s_p^2 = \frac{SS_W}{\sum_{i=1}^{I}(n_i - 1)},$$

  which is is unbiased. The subscript p stands for pooled.

- Estimates of σ^2 from the I treatments are pooled together, since SS_W can be written as

$$SS_W = \sum_{i=1}^{I} s_i^2,$$

  where s_i^2 is the sample variance in the ith group.

Estimation of Variance, continued

- Under $H_0 : \theta_1 = \ldots = \theta_I$, then the expectation of $SS_B/(I-1)$ is also σ^2.
- Thus,in this case, $SS_W/[\sum_{i=1}^{I}(n_i - 1)]$ and $SS_B/(I-1)$ should be about equal.
- If some of the τ_i are nonzero, SS_B will be inflated.
- We next develop a method of comparing the two sums of squares to find a test statistic for testing the null hypothesis that all the τ_i are equal.
- Under the assumption that the errors are normally distributed, the probability distributions of the sums of squares can be calculated.

Theorem B

- Without any loss of generality, we assume that $n_i = J$.
- If the errors, ϵ_{ij}, are independent and normally distributed with means 0 and common variances σ^2,

$$SS_W/\sigma^2 \sim \chi^2_{I(J-1)}.$$

- If, additionally, the τ_i are all equal, then SS_B/σ^2 follows a chi-square distribution with $I-1$ degrees of freedom and is independent of SS_W.

Proof of Theorem B

- Consider SS_W first.
- We first show that

$$\frac{1}{\sigma^2}\sum_{j=1}^{J}(Y_{ij}-\bar{Y}_{i.})^2$$

follow a chi-square distribution with $J-1$ degrees of freedom. There are I such sums in SS_W, and they are independent of each other since the observations are independent.

- The sum of I independent chi-square random variables that each has $J-1$ degrees of freedom follows a chi-square distribution with $I(J-1)$ degrees of freedom.

Proof of Theorem B

- We next prove that the two sums of squares are independent of each other.
- SS_W is a function of the vector U, which has elements $Y_{ij} - \bar{Y}_{i.}$, where $i = 1, \ldots, I$ and $j = 1, \ldots, J$.
- SS_B is a function of the vector V, whose elements are $\bar{Y}_{i.}$, where $i = 1, \ldots, I$ since $\bar{Y}_{..}$ can be obtained from the $\bar{Y}_{i.}$.
- Thus, it is sufficient to show that these two vectors are independent of each other. First, if $i \neq i'$, $Y_{ij} - \bar{Y}_{i.}$ and $\bar{Y}_{i'.}$ are independent since they are functions of different observations.
- Second, we can show that $Y_{ij} - \bar{Y}_{i.}$ and $\bar{Y}_{i.}$ are independent because it is easy to show $cov(Y_{ij} - \bar{Y}_{i.}, \bar{Y}_{i.}) = 0$. This completes the proof of the theorem.

Theorem C

- The statistic

$$F = \frac{SS_B/(I-1)}{SS_W/(I(J-1))}$$

  is used to test the null hypothesis $H_0 : \theta_1 = \ldots = \theta_I$.

- Under the assumption that the errors are normally distributed, the null distribution of F is the F distribution with $(I-1)$ and $I(J-1)$ degrees of freedom.
- P-value: $P(X > F)$, where X has $F(I-1, N-I)$ distribution.
- Percentage points of the F distribution are widely tabled. It can show that, under the normality assumption, the F test is equivalent to the likelihood ratio test.

Oneway ANOVA Table

Source of variation	Degrees of freedom	Sum of squares	Mean squares	F statistic
Between treatment groups	$k-1$	SSB= $\sum n_i(\bar{y}_{i.}-\bar{y})^2$	MSB= $\frac{SSB}{k-1}$	$F=\frac{MSB}{MSW}$
Within treatment groups.	$N-k$	SSW= $\sum(y_{ij}-\bar{y}_{i.})^2$	MSW= $\frac{SSW}{N-k}$	
Total.	$N-1$	SST= $\sum\sum(y_{ij}-\bar{y})^2$		

Contrasts

- Let $t = (t_1, \ldots, t_k)$ be a set of variables, either parameters or statistics, and let $a = (a_1, \ldots, a_k)$ be known constants. The function

$$\sum_{i=1}^{k} a_i t_i$$

  is called a contrast if $\sum a_i = 0$. Let $\mathcal{A} = \{a = (a_1, \ldots, a_k) : \sum_{i=1}^{k} a_i = 0\}$.

- Theorem: Let $\theta = (\theta_1, \ldots, \theta_k)$ be parameters. Then,

$$\theta_1 = \ldots = \theta_k \Longleftrightarrow \sum_{i=1}^{k} a_i \theta_i = 0 \text{ for all } a_i \text{ with } \sum_{i=1}^{k} a_i = 0$$

Another ANOVA Test

- $H_0 : \theta_1 = \ldots = \theta_k$ versus $H_1 : \theta_i \neq \theta_j$ for some $i \neq j$.
- Or, equivalently

$$H_0 : \sum_{i=1}^{k} a_i \theta_i = 0 \text{ for all } (a_1, \ldots, a_k) \text{ such that } \sum_{i=1}^{k} a_i = 0$$

versus

$$H_1 : \sum_{i=1}^{k} a_i \theta_i \neq 0 \text{ for some} (a_1, \ldots, a_k) \text{ such that } \sum_{i=1}^{k} a_i = 0$$

- For a fixed a, we define $\Omega_a = \{\theta : \sum_{i=1}^{k} a_i \theta_i = 0\}$. Then, we define the null and alternative hypothesis: $H_{0a} : \theta \in \Omega_a$ versus $H_{1a} : \theta \notin \Omega_a$
- Hence, we can write the null hypothesis as

$$H_0 : \theta \in \cap_{a \in \mathcal{A}}$$

- Based on the union-intersection methodology, we would reject H_0 if we reject H_{0a} for all $a \in \mathcal{A}$.

Another ANOVA Test

- Now let us consider to test H_{0a}, note that $Y_{ij} \sim N(\theta_i, \sigma^2)$. Define $\bar{Y}_{i.} = \frac{1}{n_i}\sum_{j=1}^{n_i} Y_{ij}$, which has $\bar{Y}_{i.} \sim N(\theta_i, \sigma^2/n_i)$. Hence

$$\frac{\sum_{i=1} a_i \bar{Y}_{i.} - \sum_{i=1} a_i\theta_i}{\sqrt{\sigma^2 \sum_{i=1}^k a_i^2/n_i}} \sim N(0,1)$$

- We need to estimate σ^2.
- Let S_i^2 be the sample variance for the ith sample:

$$S_i^2 = \frac{1}{n_i - 1}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i.})^2.$$

Then, $(n_i - 1)S_i^2\sigma^2 \sim \chi^2_{n_i-1}$. We can combine them to improve the estimator for σ^2:

$$S_p^2 = \frac{1}{N-k}\sum_{i=1}^{k}(n_i - 1)S_i^2.$$

$$T_a = \left| \frac{\sum_{i=1} a_i \bar{Y}_{i.} - \sum_{i=1} a_i\theta_i}{\sqrt{S_p^2 \sum_{i=1}^k a_i^2/n_i}} \right|$$

Another ANOVA Test

- For testing H_{0a}, we use the t statistics as follows:

$$T_a = \left| \frac{\sum_{i=1} a_i \bar{Y}_{i.} - \sum_{i=1} a_i \theta_i}{\sqrt{S_p^2 \sum_{i=1}^k a_i^2 / n_i}} \right|$$

  We reject H_{0a} if $T_a > k$ for some k.

- Thus, the union-intersection test of the ANOVA null is to reject H_0 if $sup_a T_a > k$, where k is chosen that $P_{H_0}(sup_a T_a > k) = \alpha$.

Lemma 11.2.7

- Let $(v_1, \ldots, v_k)$ be constants and let $(c_1, \ldots, c_k)$ be positive constants.
- Let $\mathcal{A} = \{a = (a_1, \ldots, a_k) : \sum a_i = 0\}$.
- Then,

$$\max_{a \in \mathcal{A}} \frac{\left(\sum_{i=1}^k a_i v_i\right)^2}{\sum_{i=1}^k a_i^2 / c_i} = \sum_{i=1}^k c_i (v_i - \bar{v}_c)^2,$$

  where $\bar{v}_c = \sum c_i v_i / \sum c_i$. The maximum is attained at any a of the form $a_i = Kc_i(v_i - \bar{v}_c)$, where K is a nonzero constant.
- Note that the we can rewrite T_a as follows: $T_a^2 = \frac{\left(\sum_{i=1}^k a_i \bar{Y}_{i.} - \sum_{i=1}^k a_i \theta_i\right)^2}{S_p^2 \sum_{i=1}^k a_i^2 / n_i} = \frac{\left(\sum_{i=1}^k a_i \bar{U}_i\right)^2}{S_p^2 \sum_{i=1}^k a_i^2 / n_i}$, where $\bar{U}_i = \bar{Y}_{i.} - \theta_i$

Another ANOVA test

- Since S_p^2 has no effect on the maximization, we can apply Lemma 11.2.7 to the above expression to get the following theorem:
- Theorem 11.2.8. For T_a defined above, we have the following result:
  - $$\max_{a \in \mathcal{A}} T_a^2 = (\sum_{i=1}^{k} n_i((\bar{Y}_{i.} - \bar{\bar{Y}}) - (\theta_i - \bar{\theta}))^2)/S_p^2$$

    where $\bar{\bar{Y}} = \sum_{i=1}^{K} n_i \bar{Y}_{i.} / \sum_i n_i$ and $\bar{\theta} = \sum_i n_i \theta_i / \sum_i n_i$.
  - Under the ANOVA assumptions that $\theta_1 = \ldots = \theta_k$,

    $$F = \max_{a \in \mathcal{A}} T_a^2 \sim (k-1) F_{k-1, N-k},$$

    an F distribution $k-1$ and $N-k$ degrees of freedom, and $N = \sum_i n_i$.
  - We reject H_0 if $F > F_{k-1, N-k, \alpha}$.
- Homework: Show Theorem 11.2.8.

Example

- Data are from Kirchhoefer (1979), who studied the measurement of chlorpheniramine maleate in tablets.
- Measurements of composites that had nominal dosages equal to 4 mg were made by seven laboratories, each laboratory making 10 measurements.
- Data is shown in the following table.
- There are two possible sources of variability in the data: variability within labs and variability between labs.

Kirchhoefer's Data

Lab 1	4.13	4.07	4.04	4.07	4.05	4.04	4.02	4.06	4.10	4.04
Lab 2	3.86	3.85	4.08	4.11	4.08	4.01	4.02	4.04	3.97	3.95
Lab 3	4.00	4.02	4.01	4.01	4.04	3.99	4.03	3.97	3.98	3.98
Lab 4	3.88	3.88	3.91	3.95	3.92	3.97	3.92	3.90	3.97	3.90
Lab 5	4.02	3.95	4.02	3.89	3.91	4.01	3.89	3.89	3.99	4.00
Lab 6	4.02	3.86	3.96	3.97	4.00	3.82	3.98	3.99	4.02	3.93
Lab 7	4.00	4.02	4.03	4.04	4.10	3.81	3.91	3.96	4.05	4.06

Analysis of Variance Table

Source	df	SS	MS	F
Labs	6	0.125	0.021	5.66
Error	63	0.231	0.0037	
Total	69	0.356		

- SS_W is the sum of squares due to error, and SS_B is the sum of squares due to labs.
- MS stands for mean square and equals the sum of squares divided by the degrees of freedom.
- The F statistic for testing the null hypothesis that there is no systematic difference among the seven labs. The F statistic has 6 and 63 df and a value of 5.66. We may thus conclude that the means of the measurements from the various labs are significantly different.

Summary of the ANOVA model

- We review the basic assumptions of the ANOVA model and comment on their importance.

$$Y_{ij} = \theta_i + \epsilon_{ij}.$$

We assume the following:

  - The ϵ_{ij} are normally distributed. 1. The F test, like the t test, remains approximately valid for moderate to large samples from moderately nonnormal distributions.
  - 2. The error variance, σ^2, is constant. In many applications, the error variances may be different in different groups. Fortunately, if there are an equal number of observations in each group, the F test is not strongly affected.
  - 3. The ϵ_{ij} are independent.This assumption is very important, both for normal theory and for the nonparametric analysis we will present later.

The Problem of Multiple Comparisons

- The application of the F test in Example above is not very informative. We concluded that the means of measurements from different labs are not all equal, but the test gives no information about how they differ, in particular about which pairs are significantly different.
- In many applications, real interest may be focused on comparing pairs or groups of treatments and estimating the treatment means and their differences.
- A naive approach would be to compare all pairs of treatment means using t tests. The difficulty with such a procedure was that although each individual comparison would have a type I error rate of α, the collection of all comparisons considered simultaneously would not.

Why is multiple testing a problem?

- Consider a case where you have 20 hypotheses to test ($H_{0i}, i = 1, \ldots, 20$), and each having a significance level of 0.05.
- What's the probability of observing at least one significant result just due to chance?
- $P($ at least one significant result $\mid H_{0i}, i = 1, \ldots, 20) = 1 - P($ no significant results $\mid H_{0i}, i = 1, \ldots, 20)$.
- Assuming all tests are independently done, we obtain that $1 - P($ no significant results $\mid H_{0i}, i = 1, \ldots, 20) = 1 - \cap_{i=1}^{20} P(ith \text{ test is no significant} \mid H_{0i}) = 1 - (1-0.05)^{20} \approx 0.64$
- So, with 20 tests being considered, we have a 64% chance of observing at least one significant result, even if all of the tests are actually not significant.

Bonferroni Correction

- Bonferroni inequality:

$$P(\cap_{i=1}^{k} A_i) \geq 1 - \sum_{i=1}^{k} (1 - P(A_i))$$

- Applying Bonferroni inequality, we obtain that $P($ at least one significant result $\mid H_{0i}, i = 1, \ldots, k) \leq \sum_{i=1}^{k} P($ ith test result is significant $\mid H_{0i})$.
- Note that $P($ ith test result is significant $\mid H_{0i})$ is the Type I error rate for the ith test. If we set the type I error rate of the ith test at α/k, then the family-wide type I error rate has the following result: $P($ at least one significant result $\mid H_{0i}, i = 1, \ldots, k) \leq \alpha$.

Bonferroni Correction

- The Bonferroni correction sets the significance cut-off at α/n.
- If k null hypotheses are to be tested, a desired overall type I error rate of at most α can be guaranteed by testing each null hypothesis at level α/k.
- Equivalently, if k confidence intervals are each formed to have confidence level $100(1-\alpha/k)\%$, they all hold simultaneously with confidence level at least $100(1-\alpha)\%$.
- The Bonferroni correction tends to be a bit too conservative because it assumes that all tests are independent of each other.

Scheffe Method

- An alternative and quite elegant approach to simultaneous inference is given by Scheffe, also called the S method.
- Under the ANOVA assumption, using the results in Theorem 11.2.8, we obtain that $F = max_a T_a^2 \sim (k-1)F_{k-1,N-k}$. Let $F_{k-1,N-k,\alpha}$ be the upper $100\alpha\%$ percentile of F distribution with $k-1$ and $N-k$ degrees of freedom.
- Let $M = \sqrt{(k-1)F_{k-1,N-k,\alpha}}$, then the probability is $1-\alpha$ that

$$\sum_{i=1}^{k} a_i \bar{Y}_{i\cdot} - M\sqrt{s_p^2 \sum_{i=1}^{k} \frac{a_i^2}{n_i}} \leq \sum_{i=1}^{k} a_i\theta_i \leq \sum_{i=1}^{k} a_i \bar{Y}_{i\cdot} + M\sqrt{s_p^2 \sum_{i=1}^{k} \frac{a_i^2}{n_i}}$$

simultaneously for all $a_1, \ldots, a_k$ with $\sum_{i=1}^{k} a_i = 0$, where

$$s_p = \frac{1}{N-k}\sum_{i=1}^{k}(n_i - 1)S_i^2.$$

Scheffe Method

- Scheffle method is valid for all contrasts.
- But it pays the price in the terms of the lengths of the intervals.
- In order to guarantee the simultaneous confidence level, the intervals may be too wide.
- The Scheffe intervals are always wider, sometimes, much wider, than the single contrast intervals.

Tukey's Method

John Rice.

- Tukey's method is used to construct confidence intervals for the differences of all pairs of means in such a way that the intervals simultaneously have a set coverage probability. The duality of confidence intervals and tests can then be used to determine which particular pairs are significantly different.
- If the sample sizes are all equal and the errors are normally distributed with a constant variance, the centered sample means, $\bar{Y}_{i.} - \theta_i$ are independent and normally distributed with means 0 and variances σ^2/k , which may be estimated by s_p^2/k. Tukey's method is based on the following test statistics:

$$max_{i_1,i_2} \frac{|\ (\bar{Y}_{i_1.} - \mu_{i_1}) - (\bar{Y}_{i_2.} - \mu_{i_2})\ |}{s_p/\sqrt{J}}$$

  where the maximum is taken over all pairs i_1 and i_2.
- This distribution is called the studentized range distribution with parameters I (the number of samples being compared) and $I(J-1)$ (the degrees of freedom in S_p).

Tukey's Method, continued

- Let the upper $100\alpha\%$ percentage point of the the studentized range distribution is denoted by $q_{I:I(J-1)}(\alpha)$.
- The simultaneously for all differences $\mu_{i_1} - \mu_{i_2}$ with confidence level $(1-\alpha)\%$ are:

$$(\bar{Y}_{i_1\cdot} - \bar{Y}_{i_2\cdot}) \pm q_{I:I(J-1)}(\alpha)\frac{s_p}{\sqrt{J}}$$

Tukey's Method, continued

- By the duality of confidence intervals and hypothesis tests, if the $100(1-\alpha)\%$ confidence interval for $(\bar{Y}_{i_1\cdot} - \bar{Y}_{i_2\cdot})$ does not include zero; that is, if

$$| \bar{Y}_{i_1\cdot} - \bar{Y}_{i_2\cdot} | > q_{I:I(J-1)}(\alpha)\frac{s_p}{\sqrt{J}},$$

  the null hypothesis that there is no difference between μ_{i_1} and μ_{i_2} may be rejected at level α.
- Also, all such hypothesis tests considered collectively have level α

Non-parametric method - The Kruskal-Wallis test

- The Kruskal-Wallis test is a generalization of the Mann-Whitney test, a non-parametric test, which can be used when normality assumption in the ANOVA model does not hold.
- The goal of the Kruskal–Wallis test is for the following null and alternative hypotheses:

$$H_0: \text{ There is no difference among the k population distributions}$$

Versus

$H_1:$ At least one of the population distributions different from the other distributions

Non-parametric method - The Kruskal-Wallis test

- Let R_{ij} be the rank of Y_{ij} in the combined sample, and let $\bar{R}_{i.} = \frac{1}{n_i}\sum_{j=1}^{J_i} R_{ij}$, the average rank in the ith group. Let

$$\bar{R}_{..} = \frac{1}{N}\sum_{i=1}^{I}\sum_{j=1}^{n_i} R_{ij} = \frac{N+1}{2},$$

  where N is the total number of observations.

- Let

$$SS_B = \sum_{i=1}^{I} n_i(\bar{R}_{i.} - \bar{R}_{..})^2$$

  which is a measure of the dispersion of the $\bar{R}_{i.}$, and $\bar{R}_{..} = \sum_{i=1}^{I} \bar{R}_{i.}/I$. SS_B can be used to test the null hypothesis that the probability distribution generating the observations under the various treatments are identical. The larger SS_B is, the stronger is the evidence against the null hypothesis.

- The exact null distribution of this statistic for various combinations of I and n_i can be enumerated, as for the Mann-Whitney test.

Non-parametric method - The Kruskal-Wallis test

- Under the null hypothesis that the probability distributions of the I groups are identical,

$$K = \frac{12}{N(N+1)} SS_B$$

is approximately distributed as a chi-squared random variable with $I-1$ degrees of freedom.

- It can also be shown that

$$K = \frac{12}{N(N+1)} (\sum_{i=1}^{I} n_i \bar{R}_{i.}^2) - 3(N+1).$$

- $$P - value = P(\chi_{I-1}^2 > K)$$

χ^2 Distribution Percentile Table

TABLE 3 Percentiles of the χ^2 Distribution—Values of χ^2_P Corresponding to P



df	$\chi^2_{.005}$	$\chi^2_{.01}$	$\chi^2_{.025}$	$\chi^2_{.05}$	$\chi^2_{.10}$	$\chi^2_{.90}$	$\chi^2_{.95}$	$\chi^2_{.975}$	$\chi^2_{.99}$	$\chi^2_{.995}$
1	.000039	.00016	.00098	.0039	.0158	2.71	3.84	5.02	6.63	7.88
2	.0100	.0201	.0506	.1026	.2107	4.61	5.99	7.38	9.21	10.60
3	.0717	.115	.216	.352	.584	6.25	7.81	9.35	11.34	12.84
4	.207	.297	.484	.711	1.064	7.78	9.49	11.14	13.28	14.86
5	.412	.554	.831	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	.676	.872	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	.989	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.73	26.76
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58	32.80

Kirchhoefer's Data Example

- For Kirchhoefer's Data above, we got $K = 29.51$.
- Referring to Chi-square Table with 6 df, we see that the p-value is less than .005 (=1-0.995).
- The nonparametric analysis also concludes that that there is a systematic difference among the 7 labs